

Complete Solution: Equation and Distance of Parallel Lines

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Question:

Find the equation of a line passing through the point $(2, 3, 2)$ and parallel to the line $\mathbf{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$. Also, find the distance between these two lines.

Solution:

The problem is solved in two parts. All vectors are represented using the `\myvec` command.

Part 1: Equation of the Required Line

The vector equation of a line is $\mathbf{r} = \mathbf{a} + \mu\mathbf{b}$, where $\mathbf{a}$ is the position vector of a point on the line and $\mathbf{b}$ is the direction vector of the line.

- 1) **Position Vector ($\mathbf{a}$):** The line passes through the point $(2, 3, 2)$. We can write its position vector as:

$$\mathbf{a} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad (1)$$

- 2) **Direction Vector ($\mathbf{b}$):** The required line is parallel to the given line, so they share the same direction vector. From the given equation, the direction vector is:

$$\mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (2)$$

Combining these gives the equation of the required line:

$$\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (3)$$

Part 2: Distance Between the Lines

The distance is found by identifying points $\mathbf{P} = \mathbf{a} + p_1\mathbf{d}$ and $\mathbf{Q} = \mathbf{b} + p_2\mathbf{d}$ on each line such that the vector $\mathbf{Q} - \mathbf{P}$ is orthogonal to the direction $\mathbf{d}$.

1. *Define Vectors:* The problem defines the two parallel lines. We can represent them with the following vectors:

$$\mathbf{a} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} \quad (\text{Point on Line 1}) \quad (4)$$

$$\mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad (\text{Point on Line 2}) \quad (5)$$

$$\mathbf{d} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (\text{Common Direction}) \quad (6)$$

2. *Solve for Parameters:* The orthogonality condition is $\mathbf{d}^\top(\mathbf{Q} - \mathbf{P}) = 0$. Expanding this gives the equation for the parameters:

$$(\mathbf{d}^\top \mathbf{d})(p_2 - p_1) + \mathbf{d}^\top(\mathbf{b} - \mathbf{a}) = 0 \quad (7)$$

We calculate the scalar products:

$$\mathbf{d}^\top \mathbf{d} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^\top \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = 4 + 9 + 36 = 49 \quad (8)$$

$$\begin{aligned} \mathbf{d}^\top(\mathbf{b} - \mathbf{a}) &= \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^\top \left(\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} \right) \\ &= \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}^\top \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} = 8 + 0 + 12 = 20 \end{aligned} \quad (9)$$

Substituting these values back:

$$49(p_2 - p_1) + 20 = 0 \quad (10)$$

$$p_2 - p_1 = -\frac{20}{49} \quad (11)$$

$$\implies p_1 - p_2 = \frac{20}{49} \quad (12)$$

We need one pair of values that satisfy this. We can simply take $\mathbf{p}_2 = \mathbf{0}$, which gives $\mathbf{p}_1 = \mathbf{20/49}$.

3. *Calculate the Distance:* Using these parameter values, we find the specific points **P** and **Q**.

$$\mathbf{P} = \mathbf{a} + p_1 \mathbf{d} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} + \frac{20}{49} \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} -58/49 \\ 87/49 \\ 120/49 \end{pmatrix} \quad (13)$$

$$\mathbf{Q} = \mathbf{b} + p_2 \mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \quad (14)$$

The vector connecting these points is:

$$\mathbf{P} - \mathbf{Q} = \begin{pmatrix} -58/49 - 98/49 \\ 87/49 - 147/49 \\ 120/49 - 98/49 \end{pmatrix} = \frac{1}{49} \begin{pmatrix} -156 \\ -60 \\ 22 \end{pmatrix} \quad (15)$$

The distance is the norm of this vector, $\|\mathbf{P} - \mathbf{Q}\|$.

$$\begin{aligned} d^2 &= \left\| \frac{1}{49} \begin{pmatrix} -156 \\ -60 \\ 22 \end{pmatrix} \right\|^2 \\ &= \left(\frac{1}{49} \right)^2 \left((-156)^2 + (-60)^2 + (22)^2 \right) \\ &= \frac{1}{2401} (24336 + 3600 + 484) = \frac{28420}{2401} = \frac{580}{49} \end{aligned} \quad (16)$$

$$d = \sqrt{\frac{580}{49}} = \frac{\sqrt{4 \times 145}}{7} \quad (17)$$

The final distance is:

$$\Rightarrow d = \boxed{\frac{2\sqrt{145}}{7}} \quad (18)$$

